



**UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION**

COURSE	:	WEB TECHNOLOGY AND APPLICATION
COURSE CODE	:	ICT600
EXAMINATION	:	JULY 2026
TIME	:	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts : PART A (5 Questions)
PART B (2 Questions)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of :
 - i) the Question Paper
 - ii) an Answer Booklet – provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages

PART A

QUESTION 1

a) Define the following terms.

- URL
- HTTP
- HTTPS
- Web Application

(8 marks)

b) Web applications can be categorized into several categories based on their functionality and the way they deliver content and services to users. Identify **FIVE (5)** categories of Web Applications.

(10 marks)

QUESTION 2

a) Define the following terms.

- JavaScript
- HTML Element
- HTML Attribute

(6 marks)

b) State **THREE (3)** HTML formatting elements used to format text on a web page.

(6 marks)

QUESTION 3

a) A library management system is required to store information about books.

Prepare an SQL statement to create a table named library_books with the following fields:

`book_id, title, author, year published`

(8 marks)

b) An online store has a table named products with the following fields:

`product_id, product_name, category, price`

Prepare SQL queries for a products table to perform the following tasks:

- Retrieve all records from the table.
- Retrieve only the product_name and price columns.
- Retrieve products with a price greater than 50.
- Retrieve products belonging to the 'Electronics' category.
- Retrieve all products sorted by price in descending order.

(10 marks)

QUESTION 4

- a) Discover **FOUR (4)** examples of popular web application frameworks. (4 marks)
- b) A startup company plans to develop a simple corporate profile website. One developer proposes using a web application framework, while another developer recommends developing the system without a framework due to the small scale of the project.

Based on this situation, demonstrate your understanding by explaining **THREE (3)** challenges faced when using a framework for this project.

(6 marks)

QUESTION 5

- a) A university web portal enables students to log in and access their personal information. However, attackers attempt to exploit the login form by submitting malicious input in the username or password field.

Classify **THREE (3)** common web application vulnerabilities that attackers may exploit in web applications.

(6 marks)

- b) A user receives an email with a suspicious attachment claiming to contain important document. After opening the file, the user's computer begins to display usual behaviour and unwanted advertisements.

Based on this situation, assess **THREE (3)** types of malicious software that may infect a computer system.

(6 marks)

PART B**QUESTION 1**

You are required to develop a web-based application for a Body Mass Index (BMI) Calculator using HTML and JavaScript.

The application must include the following features:

An HTML interface that allows users to enter:

- Weight (in kilograms)
- Height (in meters)

JavaScript functionality to:

- calculate the BMI value
- display the BMI result on the webpage
- classify the BMI result into categories (Underweight, Normal, Overweight, Obese)
- show a warning message if input is empty or invalid

(15 marks)

QUESTION 2

Construct a simple PHP web application to implement a Student Registration Form

The system should perform the following tasks:

1. Create an HTML form to collect the following information:
 - Student Name
 - Student ID
 - Gender
 - Programme
2. All fields must be required before the form can be submitted.
3. The form must use the POST method to submit the data.
4. The PHP program should:
 - Retrieve the submitted values using `$_POST`.
 - Check that all fields are filled.
 - If any field is empty, display unsuccessful message.
 - Otherwise display the submitted information in the following format:

Student Registration Details

Name: Ali Ahmad
Student ID: 2023123456
Gender: Male
Programme: Bachelor of Information
Technology

(15 marks)

END OF QUESTION PAPER